Project Report

ISE 5424 - Simulation I
Virginia Polytechnic Institute & State University

Abhiram Karri, Ashutosh Srivastava, Francisco X. Zambrano

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Abstract

This project involves estimating the operational costs of opening healthcare clinics across the U.S. for FirstTreatment, Inc. The primary goal is staffing abiding to given legal requirements for timely treatment and the need to balance staffing levels for efficiency and cost-effectiveness. The facility design consists of five stations, each with specific staffing requirements, specific to different stages of patient care. The patient flow is modeled based on statistical distributions for contact times, and the project aims to explore staffing options for various patient loads and trauma scenarios. Key parameters include the percentage of trauma patients, which varies with location based on station, and the average daily patient load, expected to range from 75 to 225 patients.

1 Purpose of the Project:

The purpose of this project is to provide FirstTreatment, Inc. with a comprehensive analysis of the staffing requirements and associated operational costs for their proposed healthcare clinics. The primary goal is to optimize staffing levels to ensure timely and efficient patient care while managing the financial aspects of operating the facilities. Considering the distinct scenarios of patient loads and trauma percentages, the project aims to help FirstTreatment make decisions on resource allocation, balancing the need for quick stabilization of trauma patients with the overall cost-effectiveness of clinic operations.

2 Executive Summary:

A simulation model was developed based on statistical data input and required outputs of staffing levels aiming for cost effectiveness and optimal staffing policy. This model was constructed using Microsoft Excel using VBASim.

We developed a simulation model in which a user can enter the number of staff for each of the 5 stations, the load and the percentage of the trauma patients. The model will take these inputs and show the average waiting times for patients in front of each station for 1000 replications. The Output tables shows the Mean of all these average waiting times at each station along with the Confidence intervals half width of normal and t distributions. Since there are in total four different combinations of load and trauma patients' percentage, we ran the model for all these combinations and the results of these simulations are summarized in the tables in the analysis section.

3 Simulation Design and Analysis:

The simulation of the Clinic was designed by the information provided. Incoming patients were modeled using a homogeneous Poisson process with an inter arrival rate of 14.4 or 4.8 minutes. All the patients will Sign In station first, contact times for Sign In station is Exponential distribution with a mean of 3 minutes. It is provided that out of all the patients visiting the facility in a day, 8%-12% are Trauma patients depending on where the facility is located. Depending upon the condition of the patient, he/she will go to the next station. In case of trauma patients, they will be served by a trauma doctor in the trauma station with Exponential distribution with 90 minutes mean and after that the trauma patients will enter the treatment station where they will be served with lognormal distribution with mean 30 minutes and variance 4 minutes. In case of non-trauma patients, they will go to the registration station where they will be served with Lognormal distribution with mean 5 minutes and variance 2 minutes. The non-trauma patients will further go to the Examination station where the contact time is distributed as Normal with mean of 16 minutes and variance of 3 minutes. And lastly the non-trauma patients will enter the treatment station where they will be served with lognormal distribution with mean 13.3 minutes and variance 2 minutes. Patients were routed to their respective station staff and a First-In-First-Out (FIFO) service strategy was assumed. When a staff member of a particular station is not available, the patient is placed into a FIFO queue. There are in total of five FIFO queues, one in front

of each station. Only the Treatment station queue has both trauma and non-trauma patients.

For the current system, simulation model build is run for different patient loads ranging from lowest 75 to highest 225 with percentage of trauma patients being 8%-12%.
We run the simulation for 1000 replications, The widths of the confidence intervals indicate that this number of simulation replications provides sufficient precision (see Table 1 and 2)

Output	Avg Wait Sign In	Avg Wait Registration	Avg Wait Examination	Avg Wait Trauma	Avg Wait Treatment
Mean	0.79	1.84	3.03	3.50	2.62
95% CI half width (normal)	0.025231206	0.040079863	0.080500334	0.696970273	0.159539054
95% CI half width (t)	0.03	0.046214536	0.092821814	0.803649397	0.183958297

Figure 1: Table1: Patient load 75 and trauma-8%

Simulation was run for patient load of 75, with percentage of trauma patients being 8%. The optimal staffing policy turned out to be: SignIn station (1), Registration(1), Examination(2), Trauma(2), Treatment(2). [Station(No.of Staff)]

Output	Avg Wait Sign In	Avg Wait Registration	Avg Wait Examination	Avg Wait Trauma	Avg Wait Treatment
Mean	0.33	1.50	4.82	2.31	15.57
95% CI half width (normal)	0.009896163	0.027141688	0.119915295	0.324522554	0.376773702
95% CI half width (t)	0.01	0.031296028	0.138269677	0.374194374	0.434443146

Figure 2: Table2: Patient load 225 and trauma-12%

Simulation was run for patient load of 225, with percentage of trauma patients being 12%. The optimal staffing policy turned out to be: SignIn station (2), Registration(2), Examination(4), Trauma(5), Treatment(3). [Station(No.of Staff)]

4 Input Modeling and Simulation Structure

FirstTreatment, Inc. provided the input for the model. To summarize, patient arrivals were modeled using a homogenous Poisson process with a load of 75 to 225 patients per day, using implies 14.4 minutes of interarrival time for 75 patients load and 4.8 minutes of interarrival time for 225 patients load respectively. Contact times for Sign In station was modeled using Exponential distribution with mean 3. Contact times for Registration station was modeled using Lognormal distribution with mean 5 and variance 2. Contact times for Examination station was modeled using normal distribution with mean 16 and variance 3. Contact times for Trauma station was modeled using Exponential distribution with mean 90. Contact times for Treatment station was modeled using Lognormal distribution with mean 30 and variance 4 for trauma patients and Lognormal distribution with mean 13.3 and variance 2 for non-trauma patients.

Input Model	Distribution	Mean	Var	Seed
Sign In	Exponential	3 min	-	2
Registration	Lognormal	5 min	2 min	5
Examination	Normal	16 min	3 min	6
Trauma	Exponential	90 min	-	4
Treatment trauma Patients	Lognormal	30 min	4 min	7
Treatment non trauma Patients	Lognormal	13.3 min	2 min	8

Figure 3: Table: Input Modelling

Figure 5 and 6 in the Appendix illustrate the structures of the simulation of the clinic system.

5 Conclusion

To summarize, we found that for the heaviest load, that is 225 patients and 12% Trauma patients, we need 2 staffs at Sign In station, 2 staffs at Registration station, 4 staffs at Examination station, 5 staffs at Trauma station and 3 staffs at Treatment station. Similarly we found that for the lightest load, that is 75 patients and 8% Trauma patients, we need 1 staffs at Sign In station, 1 staffs at Registration station, 2 staffs at Examination station, 2 staffs at Trauma station and 2 staffs at Treatment station. We also concluded that for the two scenarios of (225, 12%) and (225,8%) the staffing level at each of the stations will be exactly same, except for the Trauma station, where we need one more staff in the 12% case as compared to the 8% case. Similarly, for the two scenarios of (75, 12%) and (75,8%) the staffing level at each of the stations will be exactly same, except for the Trauma station, where we need one more staff in the 12% case as compared to the 8% case.

Summary Table

Stations>	Sign In	Registration	Examination	Trauma	Treatment
225, 12%	2	2	4	5	3
225, 8%	2	2	4	4	3
75, 12%	1	1	2	3	2
75, 8%	1	1	2	2	2

Figure 4: Table: Summary

Providing insight on whether slightly relaxing the service requirements might reduce staffing costs, let's take the example of the heaviest load, that is 225 patients and 12% Trauma patients. We analyzed that if we reduce one staff at the Sign In center, we can meet the service requirement as the Mean average waiting time at Sign station will go from 0.03 minutes to 4.86 minutes, but in that case the Sign In won't be "very fast". We have also seen from the model that if we reduce the Examination staff by 1, the waiting time will go from 5 minutes to 30 minutes. As we are allowed to have a waiting time of 15 to 20 minutes at this station. This shows that we are compromising the service requirement for this station by 10 minutes If we reduce the staffing cost.

6 Appendix

1. **First Treatment Inc. Clinic**

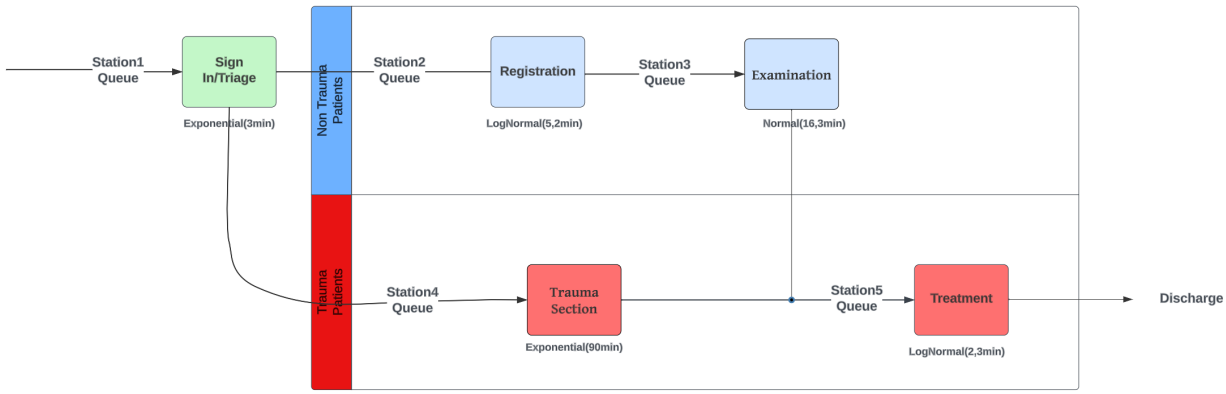

Figure 5: First Inc Clinic Flow Chart

2. **Simulation Flow Chart for Simulation of Current model.**

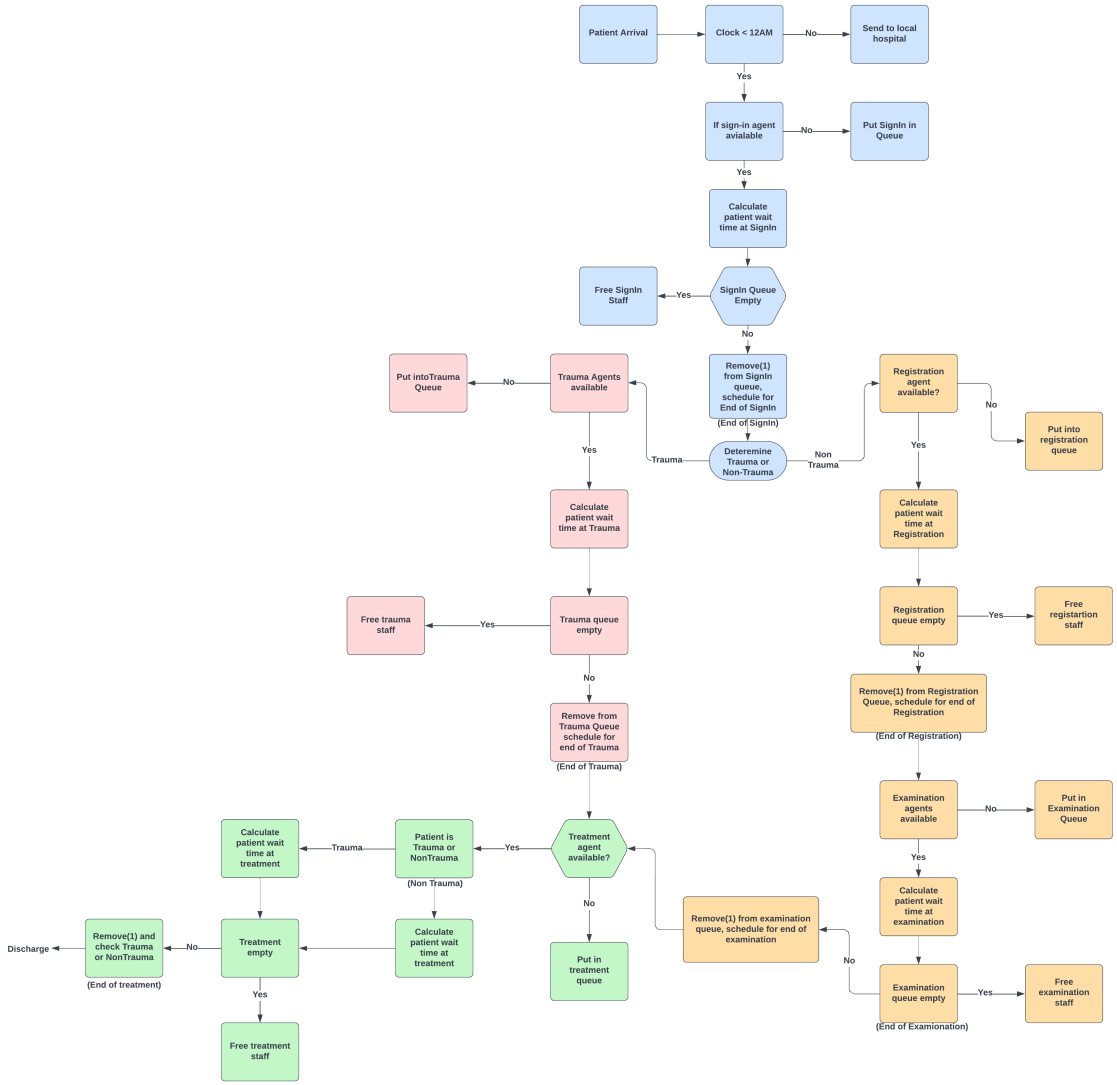

Figure 6: Simulation Flow Chart

3. **Output for 4 different scenarios**

G	H	I	J	K	L
INPUT					
Station >	Sign IN	Registration	Examination	Trauma	Treatment
No. of Staffs >	1	1	2	2	2
Load >	75				
MeanArrival	14.4				
Arrival Rate >	0.069				
Trauma Percen >	8				
OUTPUT	Avg Wait SignIn	Avg Wait Regis	Avg Wait Exami	Avg Wait Traum	Avg Waiting Treat
Mean	0.79	1.84	3.03	3.50	2.62
95% CI half width (normal)	0.025231206	0.040079863	0.080500334	0.696970273	0.159539054
95% CI half width (t)	0.03	0.046214536	0.092821814	0.803649397	0.183958297

Figure 7: Output for 75 load and 8% trauma patients

G	H	I	J	K	L
INPUT					
Station >	Sign IN	Registration	Examination	Trauma	Treatment
No. of Staffs >	1	1	2	3	2
Load >	75				
MeanArrival	14.4				
Arrival Rate >	0.069				
Trauma Percen >	12				
OUTPUT	Avg Wait SignIn	Avg Wait Regis	Avg Wait Exami	Avg Wait Traum	Avg Waiting Treat
Mean	0.79	1.73	2.74	1.21	4.62
95% CI half width (normal)	0.025231206	0.040704278	0.073901351	0.283611392	0.242641287
95% CI half width (t)	0.03	0.046934525	0.085212782	0.327021299	0.27978026

Figure 8: Output for 75 load and 12% trauma patients

G	H	I	J	K	L
INPUT					
Station >	Sign IN	Registration	Examination	Trauma	Treatment
No. of Staffs >	2	2	4	4	3
Load >	225				
MeanArrival	4.8				
Arrival Rate >	0.208				
Trauma Percen >	8				
OUTPUT	Avg Wait SignIn	Avg Wait Regis	Avg Wait Exami	Avg Wait Traum	Avg Waiting Treat
Mean	0.33	1.65	5.73	1.95	9.43
95% CI half width (normal)	0.009896163	0.028253555	0.152288007	0.350178951	0.256233373
95% CI half width (t)	0.01	0.032578078	0.175597396	0.40377777	0.295452767

Figure 9: Output for 225 load and 8% trauma patients

G	H	I	J	K	L
INPUT					
Station >	Sign IN	Registration	Examination	Trauma	Treatment
No. of Staffs >	2	2	4	5	3
Load >	225				
MeanArrival	4.8				
Arrival Rate >	0.208				
Trauma Percen >	12				
OUTPUT	Avg Wait SignIn	Avg Wait Regis	Avg Wait Exami	Avg Wait Traum	Avg Waiting Treat
Mean	0.33	1.50	4.82	2.31	15.57
95% CI half width (normal)	0.009896163	0.027141688	0.119915295	0.324522554	0.376773702
95% CI half width (t)	0.01	0.031296028	0.138269677	0.374194374	0.434443146

Figure 10: Output for 225 load and 12% trauma patients

4. **VBASim Code of Simulation of Current System**

```
' Clinic Simulation
' Station1 = SignIn
' Station2 = Registration
' Station3 = Examination
' Station4 = Trauma
' Station5 = Treatment

' Parameters we may want to change

Dim MeanArrival As Double              ' mean arrival of
   patients
Dim MeanStation1 As Double           ' mean entry time station1
Dim MeanStation2 As Double           ' mean entry time station2
Dim VarStation2 As Double              ' variance entry time
   station2
Dim MeanStation3 As Double           ' mean entry time station3
Dim VarStation3 As Double              ' variance entry time
   station3
Dim MeanStation4 As Double           ' mean entry time station4
Dim MeanStation5TP As Double        ' mean entry time for Trauma
    Patients station5
Dim VarStation5TP As Double           ' variance entry time for
   Trauma Patients station5
Dim MeanStation5NTP As Double     ' mean entry time for Non
   Trauma Patients station5
Dim VarStation5NTP As Double        ' variance entry time for
   Non Trauma Patients station5

Dim RunLength As Double          ' length of the working day
Dim TraumaPerc As Double
Dim NonTraumaPercTreatment As Double


Dim NumStaffs1 As Integer         ' number of staffs at SignIn
   station1
Dim NumStaffs2 As Integer         ' number of staffs at
   Registration station2
Dim NumStaffs3 As Integer         ' number of staffs at
   Examination station3
Dim NumStaffs4 As Integer         ' number of staffs at Trauma
   station4
Dim NumStaffs5 As Integer         ' number of staffs at
   Treatment station5
```

```
'Global objects needed for simulation

Dim Station1Q As New FIFOQueue        ' queue for station1
Dim Station2Q As New FIFOQueue        ' queue for station2
Dim Station3Q As New FIFOQueue        ' queue for station3
Dim Station4Q As New FIFOQueue        ' queue for station4
Dim Station5Q As New FIFOQueue        ' queue for station5

Dim Station1Wait As New DTStat        ' discrete-time statistics
    on station1 waiting
Dim Station2Wait As New DTStat        ' discrete-time statistics
    on station2 waiting
Dim Station3Wait As New DTStat        ' discrete-time statistics
    on station3 waiting
Dim Station4Wait As New DTStat        ' discrete-time statistics
    on station4 waiting
Dim Station5Wait As New DTStat        ' discrete-time statistics
    on station5 waiting

Dim StaffS1 As New Resource           ' Station1 resource
Dim StaffS2 As New Resource           ' Station2 resource
Dim StaffS3 As New Resource           ' Station3 resource
Dim StaffS4 As New Resource           ' Station4 resource
Dim StaffS5 As New Resource           ' Station5 resource

Public Sub ClinicSim()

    Dim Reps As Integer
    Dim NextEvent As EventNotice

    ' Read in staffing policy

    NumStaffs1 = Worksheets("FTI2").Cells(3, 8)
    NumStaffs2 = Worksheets("FTI2").Cells(3, 9)
    NumStaffs3 = Worksheets("FTI2").Cells(3, 10)
    NumStaffs4 = Worksheets("FTI2").Cells(3, 11)
    NumStaffs5 = Worksheets("FTI2").Cells(3, 12)

    ' Read in Load(or Arrival rate andThe trauma patient
   Percentage

    MeanArrival = Worksheets("FTI2").Cells(6, 8)
    TraumaPerc = Worksheets("FTI2").Cells(8, 8)

    Call MyInit

    For Reps = 1 To 100
```

```
        Call VBASimInit
        StaffS1.SetUnits (NumStaffs1)
        StaffS2.SetUnits (NumStaffs2)
        StaffS3.SetUnits (NumStaffs3)
        StaffS4.SetUnits (NumStaffs4)
        StaffS5.SetUnits (NumStaffs5)

       Call Schedule("Arrival", Expon(MeanArrival, 1))

        Do
            Set NextEvent = Calendar.Remove
            Clock = NextEvent.EventTime
            Select Case NextEvent.EventType
            Case "Arrival"
                Call Arrival
            Case "EndofSignIn"
                Call EndofSignIn(NextEvent.WhichObject)
            Case "EndofRegistration"
                Call EndofRegistration(NextEvent.WhichObject)
            Case "EndofExamination"
                Call EndofExamination(NextEvent.WhichObject)
            Case "EndofTrauma"
                Call EndofTrauma(NextEvent.WhichObject)
            Case "EndofTreatment"
                Call EndofTreatment(NextEvent.WhichObject)
            End Select
        Loop Until Calendar.N = 0 ' stop when event calendar
   empty

        ' Write output report for each replication

        Call Report(Station1Wait.Mean, "FTI2", Reps + 1, 1)
        Call Report(Station2Wait.Mean, "FTI2", Reps + 1, 2)
        Call Report(Station3Wait.Mean, "FTI2", Reps + 1, 3)
        Call Report(Station4Wait.Mean, "FTI2", Reps + 1, 4)
        Call Report(Station5Wait.Mean, "FTI2", Reps + 1, 5)

    Next Reps

    End

End Sub

Private Sub Arrival()

' Schedule next Patient arrival if < 12 AM (Midnight)
    If Clock < RunLength Then
```

```
        Call Schedule("Arrival", Expon(MeanArrival, 1))
    Else
        Exit Sub
    End If

    ' Process the newly arriving Patient
    Dim Patient As New Entity
    Patient.MarkTime = Clock

    'Check If Sign In Staff is free
    If StaffS1.Busy < StaffS1.NumberOfUnits Then
        StaffS1.Seize (1)
        Station1Wait.Record (0)
        Call SchedulePlus("EndofSignIn", Expon(MeanStation1, 2)
   , Patient)
    Else
        Station1Q.Add Patient
    End If

    Set Patient = Nothing

End Sub
Private Sub EndofSignIn(DepartingPatient As Entity)

     Dim Wait As Double

    If Uniform(0, 1, 3) < TraumaPerc / 100 Then
       DepartingPatient.MarkPatient = "TRAUMATPATIENT"
       Call TraumaArrival(DepartingPatient)
    Else
       Call RegistrationArrival(DepartingPatient)
    End If

     If Station1Q.NumQueue > 0 And StaffS1.NumberOfUnits >=
   StaffS1.Busy Then
          Set DepartingPatient = Station1Q.Remove
          Wait = Clock - DepartingPatient.MarkTime
          Station1Wait.Record (Wait)
          Call SchedulePlus("EndofSignIn", Expon(MeanStation1,
   2), DepartingPatient)
          DepartingPatient.MarkTime = Clock
       Else
          StaffS1.Free (1)
       End If

       Set DepartingPatient = Nothing
```

```
End Sub
Private Sub TraumaArrival(TraumaPatient As Entity)

       'Check If Trauma Staff is free
        If StaffS4.Busy < StaffS4.NumberOfUnits Then
           StaffS4.Seize (1)
           Station4Wait.Record (0)
           Call SchedulePlus("EndofTrauma", Expon(MeanStation4,
    4), TraumaPatient)
        Else
           Station4Q.Add TraumaPatient
        End If

    Set TraumaPatient = Nothing

End Sub
Private Sub EndofTrauma(DepartingPatient As Entity)

  Dim Wait As Double

     If Station4Q.NumQueue > 0 And StaffS4.NumberOfUnits >=
   StaffS4.Busy Then
        Set DepartingPatient = Station4Q.Remove
        Wait = Clock - DepartingPatient.MarkTime
        Station4Wait.Record (Wait)
        Call SchedulePlus("EndofTrauma", Expon(MeanStation4, 4)
   , DepartingPatient)
        DepartingPatient.MarkTime = Clock
    Else
        StaffS4.Free (1)
    End If

    Call TreatmentArrival(DepartingPatient)

    Set DepartingPatient = Nothing

End Sub
Private Sub RegistrationArrival(NonTraumaPatient As Entity)

   'Check If Registration Staff is free
    If StaffS2.Busy < StaffS2.NumberOfUnits Then
        StaffS2.Seize (1)
        Station2Wait.Record (0)
        Call SchedulePlus("EndofRegistration", Lognormal(
   MeanStation2, VarStation2, 5), NonTraumaPatient)
    Else
        Station2Q.Add NonTraumaPatient
```

```
    End If

    Set NonTraumaPatient = Nothing

End Sub
Private Sub EndofRegistration(DepartingPatient As Entity)

    Dim Wait As Double

     If Station2Q.NumQueue > 0 And StaffS2.NumberOfUnits >=
   StaffS2.Busy Then
        Set DepartingPatient = Station2Q.Remove
        Wait = Clock - DepartingPatient.MarkTime
        Station2Wait.Record (Wait)
        Call SchedulePlus("EndofRegistration", Lognormal(
   MeanStation2, VarStation2, 5), DepartingPatient)
        DepartingPatient.MarkTime = Clock
    Else
        StaffS2.Free (1)
    End If

    Call ExaminationArrival(DepartingPatient)

     Set DepartingPatient = Nothing

End Sub
Private Sub ExaminationArrival(NonTraumaPatient As Entity)

    'Check If Examination Staff is free
    If StaffS3.Busy < StaffS3.NumberOfUnits Then
        StaffS3.Seize (1)
        Station3Wait.Record (0)
        Call SchedulePlus("EndofExamination", Normal(
   MeanStation3, VarStation3, 6), NonTraumaPatient)
    Else
        Station3Q.Add NonTraumaPatient
    End If

    Set NonTraumaPatient = Nothing

End Sub
Private Sub EndofExamination(DepartingPatient As Entity)

    Dim Wait As Double

     If Station3Q.NumQueue > 0 And StaffS3.NumberOfUnits >=
   StaffS3.Busy Then
```

```
        Set DepartingPatient = Station3Q.Remove
        Wait = Clock - DepartingPatient.MarkTime
        Station3Wait.Record (Wait)
        Call SchedulePlus("EndofExamination", Normal(
   MeanStation3, VarStation3, 6), DepartingPatient)
        DepartingPatient.MarkTime = Clock
     Else
        StaffS3.Free (1)
     End If

    If Uniform(0, 1, 9) < NonTraumaPercTreatment / 100 Then
        Call TreatmentArrival(DepartingPatient)
    End If

     Set DepartingPatient = Nothing

End Sub
Private Sub TreatmentArrival(Patient As Entity)

 'Check If Treatment Staff is free
 If Patient.MarkPatient = "TRAUMATPATIENT" Then
      If StaffS5.Busy < StaffS5.NumberOfUnits Then
         StaffS5.Seize (1)
         Station5Wait.Record (0)
         Call SchedulePlus("EndofTreatment", Lognormal(
   MeanStation5TP, VarStation5TP, 7), Patient)
     Else
         Station5Q.Add Patient
     End If
 Else
     If StaffS5.Busy < StaffS5.NumberOfUnits Then
         StaffS5.Seize (1)
         Station5Wait.Record (0)
         Call SchedulePlus("EndofTreatment", Lognormal(
   MeanStation5NTP, VarStation5NTP, 8), Patient)
      Else
        Station5Q.Add Patient
      End If
  End If

    Set Patient = Nothing

End Sub
Private Sub EndofTreatment(DepartingPatient As Entity)

  If DepartingPatient.MarkPatient = "TRAUMATPATIENT" Then
       If Station5Q.NumQueue > 0 And StaffS5.NumberOfUnits >=
```

```
   StaffS5.Busy Then
          Set DepartingPatient = Station5Q.Remove
          Wait = Clock - DepartingPatient.MarkTime
          Station5Wait.Record (Wait)
          Call SchedulePlus("EndofTreatment", Lognormal(
   MeanStation5TP, VarStation5TP, 7), DepartingPatient)
       Else
          StaffS5.Free (1)
       End If
    Else
        If Station5Q.NumQueue > 0 And StaffS5.NumberOfUnits >=
   StaffS5.Busy Then
           Set DepartingPatient = Station5Q.Remove
           Wait = Clock - DepartingPatient.MarkTime
           Station5Wait.Record (Wait)
           Call SchedulePlus("EndofTreatment", Lognormal(
   MeanStation5NTP, VarStation5NTP, 8), DepartingPatient)
        Else
           StaffS5.Free (1)
        End If
    End If

     Set DepartingPatient = Nothing

End Sub

Private Sub MyInit()

' Initialize the simulation
    Call InitializeRNSeed

    MeanStation1 = 3
    MeanStation2 = 5
    VarStation2 = 2#
    MeanStation3 = 16
    VarStation3 = 3#
    MeanStation4 = 90
    MeanStation5TP = 30
    VarStation5TP = 4#
    MeanStation5NTP = 13.3
    VarStation5NTP = 2#
    RunLength = 1080
    NonTraumaPercTreatment = 60

 'Add statistics that need to be
' initialized between replications to the global collections
    TheDTStats.Add Station1Wait
```

```
    TheDTStats.Add Station2Wait
    TheDTStats.Add Station3Wait
    TheDTStats.Add Station4Wait
    TheDTStats.Add Station5Wait

' Write headings for the Output reports

    Call Report("Avg Wait SignIn", "FTI2", 1, 1)
    Call Report("Avg Wait Registration", "FTI2", 1, 2)
    Call Report("Avg Wait Examination", "FTI2", 1, 3)
    Call Report("Avg Wait Trauma", "FTI2", 1, 4)
    Call Report("Avg Wait Treatment", "FTI2", 1, 5)

End Sub
```